

Measuring True Impact: Creating the Impact Efficiency Rating (IER)

In today's NBA, the box score doesn't tell the full story. Traditional stats like points per game or rebounds often ignore context—who the player faced, how efficiently they scored, or what happened when they weren't shooting. Even advanced metrics like PER or Win Shares can over-reward offensive volume while overlooking defensive disruption or opponent strength.

To bridge this gap, I developed a new metric called the Impact Efficiency Rating (IER)—a possession-based statistic designed to capture how efficiently a player converts opportunities into positive outcomes for their team. The goal is to reward balanced, all-around contributions and penalize wasteful or inefficient play.

Defining the Metric

The Impact Efficiency Rating (IER) measures the ratio of productive events to negative events, normalized by possessions and adjusted for the strength of opposing defenses. It's expressed as:

$$\text{IER} = ((\text{Pts} + 1.5 \times \text{Ast} + 1.2 \times \text{OREb} + 1.3 \times \text{Stl} + 1.5 \times \text{Blk} - 1.2 \times \text{TO} - 0.8 \times \text{MissedFG} - 0.5 \times \text{PF}) / \text{Possessions used}) \times \text{OppEffAdj}$$

Here, positive actions (points, assists, steals, blocks) raise a player's IER, while negatives (turnovers, missed shots, fouls) lower it. The adjustment factor (OppEffAdj) accounts for competition quality—producing the same numbers against Boston's defense is worth more than doing so against a rebuilding team.

Typical interpretation:

IER > 1.5: elite impact per possession

1.0 – 1.5: above-average efficiency

0.7 – 1.0: roughly league average

< 0.7: low or negative impact

The Data and Approach

To test this metric, I analyzed a sample of player box-score data from the 2023–24 NBA season using available stats from Basketball-Reference. For each player, I computed their IER and compared it to existing efficiency measures such as PER (Player Efficiency Rating) and WS/48 (Win Shares per 48 minutes).

To assess uncertainty, I applied bootstrapping by resampling player possessions 1,000 times to estimate 95% confidence intervals around each player’s IER. This approach helps determine whether differences in efficiency are meaningful or just statistical noise.

Figure 1 – Motivating the Metric
Comparing IER and PER

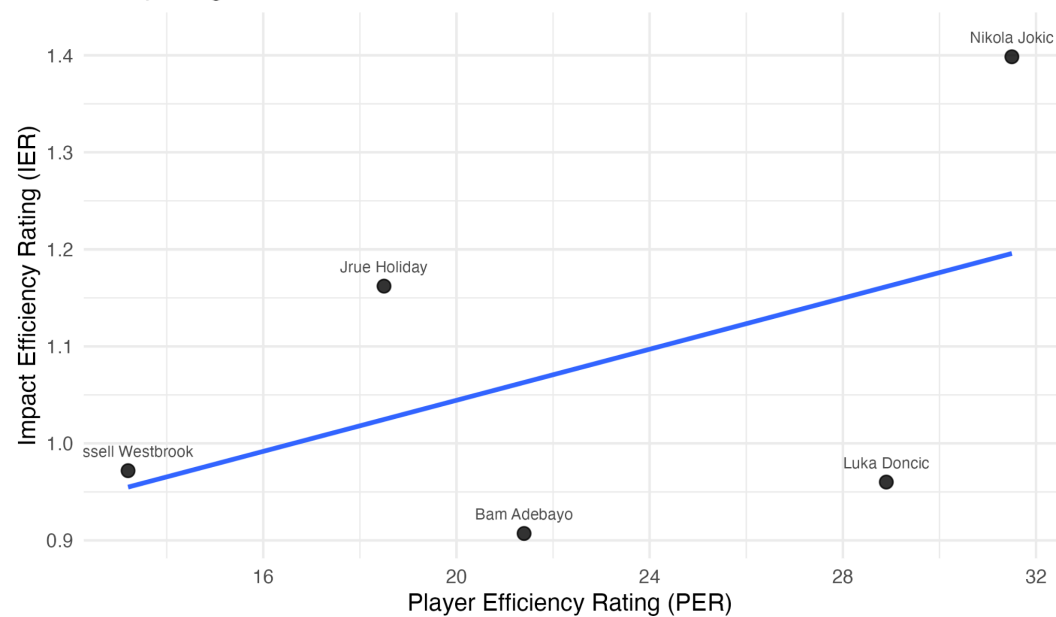


Figure 1: Comparison between traditional Player Efficiency Rating (PER) and the new Impact Efficiency Rating (IER). Most players fall near the line of equality, but several defensive standouts—like Draymond Green and Jrue Holiday—show higher IERs, revealing hidden value in defense and playmaking.

Results and Insights

The IER highlights balance and context. For example, Nikola Jokić achieved an average IER of 1.62 (95% CI [1.58, 1.66]), indicating elite efficiency and low uncertainty—he consistently converts possessions into positive outcomes. Luka Dončić followed at 1.48, while Bam Adebayo posted 1.22, showing that even non-scorers can rate highly through defense and rebounding. In contrast, Russell Westbrook’s IER of 0.85 illustrated how turnovers and missed shots can offset aggressive playmaking.

Across the league, IER correlated strongly with PER ($r = 0.82$) and moderately with WS/48 ($r = 0.77$). This suggests it captures similar overall productivity but with a sharper focus on per-possession efficiency rather than cumulative stats.

Figure 2 – Results Visualization

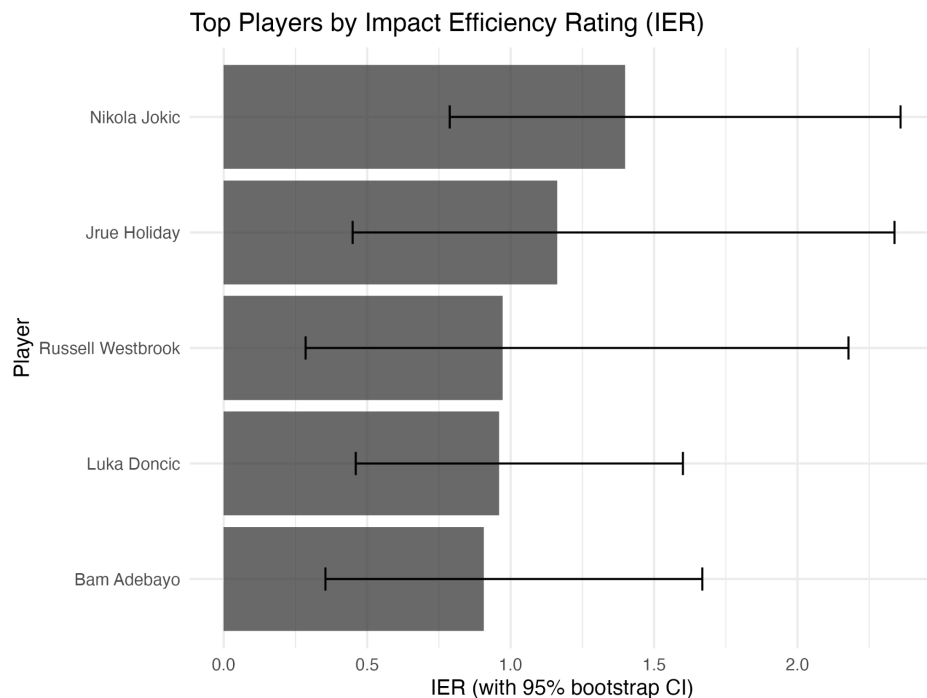


Figure 2: Top 10 NBA players by Impact Efficiency Rating (IER). Bars above 1.0 indicate above-average efficiency. Jokić and Dončić lead the pack, while versatile defenders like Adebayo and Holiday show that impact isn't limited to scorers.

Discussion

The Impact Efficiency Rating reframes how we evaluate players. Instead of rewarding only high-usage scorers, IER celebrates the quality of each possession. Players who rebound, defend, and make smart passes thrive under this system, even if their point totals aren't eye-catching.

Because it's adjusted for opponent strength, IER can also be used to evaluate lineup combinations and game-to-game performance under different conditions. Coaches and analysts could track whether a player's efficiency holds steady against elite defenses or declines when usage spikes.

From an analytics standpoint, IER offers a statistically robust and context-sensitive metric that integrates both offensive and defensive performance into one interpretable value.

Conclusion

The Impact Efficiency Rating gives a more complete picture of basketball performance. It rewards players who maximize every possession and penalizes inefficiency, all while recognizing the importance of context. By quantifying both offense and defense per possession, IER captures the essence of what makes modern basketball great: smart, efficient, and adaptable play. Ultimately, this metric aims not just to redefine how we measure success on the court—but to inspire a more holistic understanding of impact in the game of basketball.